

LAPORAN PRAKTIKUM ETS



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JAWA TIMUR
2026**

Uji Kit IMCLab Melalui Serial Arduino

Code On OFF Motor

```
int ledPin = 2;
char myData = 0;

int motor1Pin1 = 27;
int motor1Pin2 = 26;
int enable1Pin = 12;

// Setting PWM properties
const int freq = 30000;
const int pwmChannel = 0;
const int resolution = 8;
int dutyCycle = 200;

const byte pin_rpm = 13;
int volatile rev = 0;
//int rpm = 0;

void setup()
{
  pinMode(ledPin, OUTPUT);
  Serial.begin(115200);
  // sets the pins
  pinMode(motor1Pin1, OUTPUT);
  pinMode(motor1Pin2, OUTPUT);
  pinMode(enable1Pin, OUTPUT);
  pinMode(pin_rpm, INPUT_PULLUP);
  // pinMode(pin_rpm, INPUT);
  //attachInterrupt(digitalPinToInterrupt(pin_rpm), isr, RISING);

  // configure LED PWM functionalitites
  ledcSetup(pwmChannel, freq, resolution);

  // attach the channel to the GPIO to be controlled
  ledcAttachPin(enable1Pin, pwmChannel);

  // testing
  Serial.print("Testing DC Motor...");
}

void MotorOn()
{
  // Motor On - Move DC motor forward
  dutyCycle = 205;
  digitalWrite(motor1Pin1, HIGH);
  digitalWrite(motor1Pin2, LOW);
}
```

```
    ledcWrite(pwmChannel, dutyCycle);
}

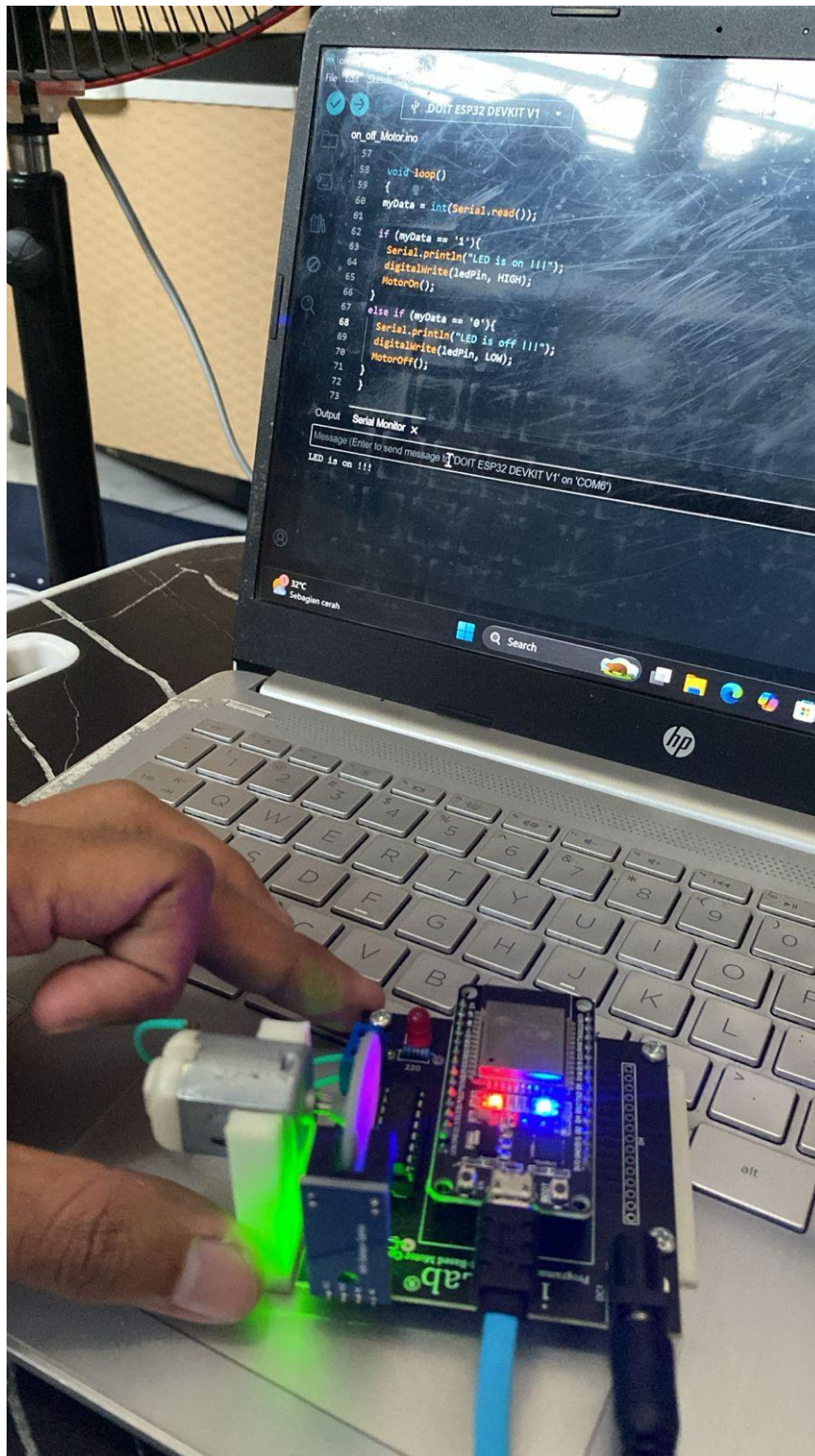
void MotorOff()
{
    dutyCycle = 0;
    digitalWrite(motor1Pin1, HIGH);
    digitalWrite(motor1Pin2, LOW);
    ledcWrite(pwmChannel, dutyCycle);
}

void loop()
{
    myData = int(Serial.read());

    if (myData == '1'){
        Serial.println("LED is on !!!");
        digitalWrite(ledPin, HIGH);
        MotorOn();
    }
    else if (myData == '0'){
        Serial.println("LED is off !!!");
        digitalWrite(ledPin, LOW);
        MotorOff();
    }
}
```

Hasil Uji IMCLab Melalui Arduino

ON



OFF



Uji Kit IMCLab Melalui Jupyter Python

Code Test Port Serial

```
import serial
#arduinoData = serial.Serial('/dev/ttyACM1', 9600)
arduinoData = serial.Serial('COM6', 115200, timeout=0)

while True:
    myData = input('Kirimkan perintah (1/0): ')
    if myData == "1":
        arduinoData.write(b'1')
    elif myData == "0":
        arduinoData.write(b'0')
```

Code Imc Lab.py

```
import serial
import serial.tools.list_ports
import time

class iMCLab:
    def __init__(self, port=None, baudrate=115200):
        # Jika port tidak diisi, program akan mencari otomatis
        if port is None:
            ports = list(serial.tools.list_ports.comports())
            for p in ports:
                # Mencari perangkat USB/Arduino/ESP32
                if 'USB' in p.description or 'CH340' in p.description or 'CP210' in p.description:
                    port = p.device
                    break

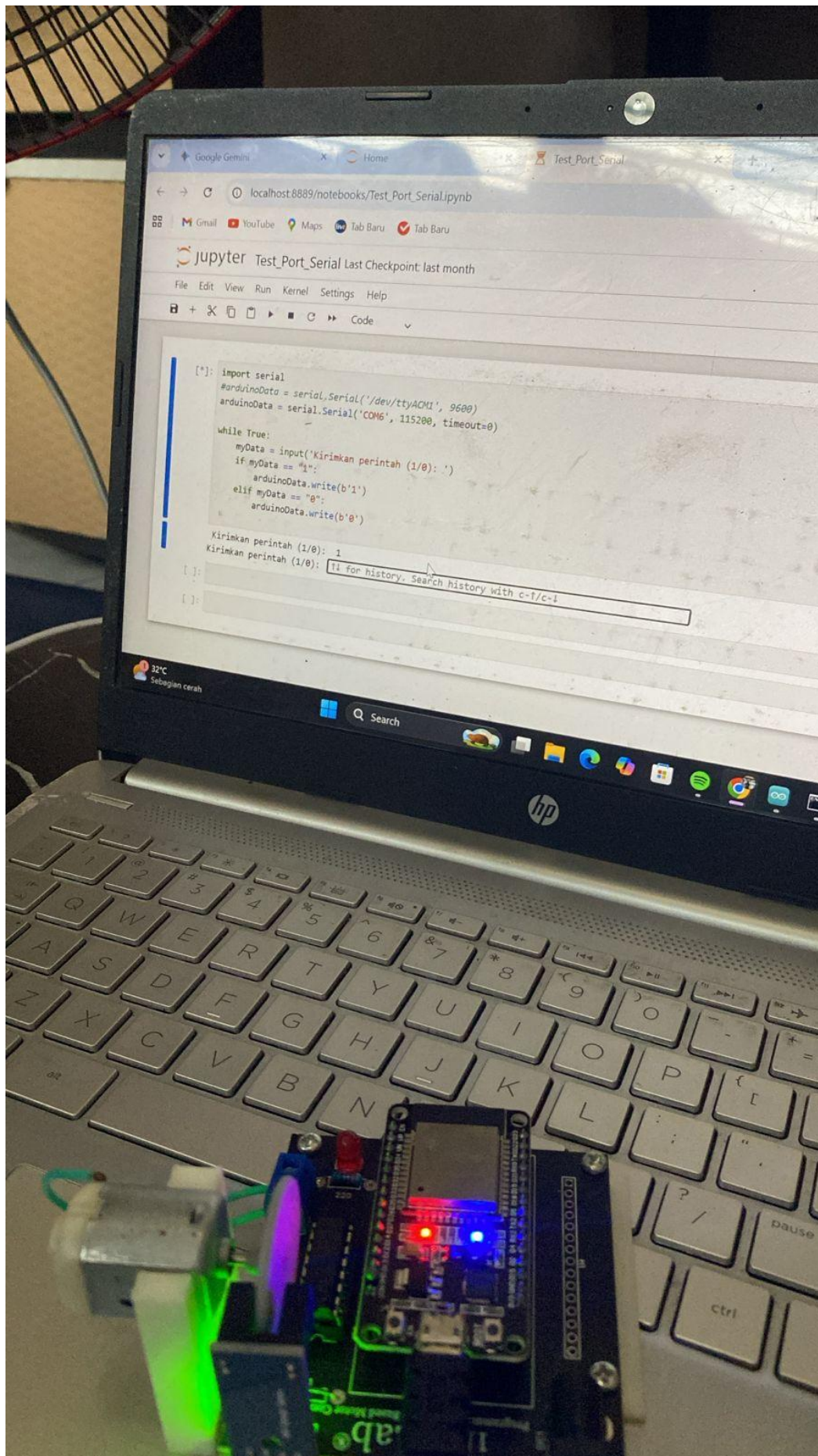
        try:
            self.ser = serial.Serial(port, baudrate, timeout=1)
            time.sleep(2) # Memberi waktu ESP32 untuk reset setelah koneksi
            print(f"Opening connection")
            print(f"iMCLab connected via Arduino on port {port}")
        except Exception as e:
            print(f"Gagal koneksi: {e}")

    def LED(self, value):
        # Mengirim perintah ke ESP32 dalam format 'LED:nilai'
        if self.ser.is_open:
            command = f"LED:{value}\n"
            self.ser.write(command.encode())
        else:
            print("Error: Koneksi serial tertutup.")
```

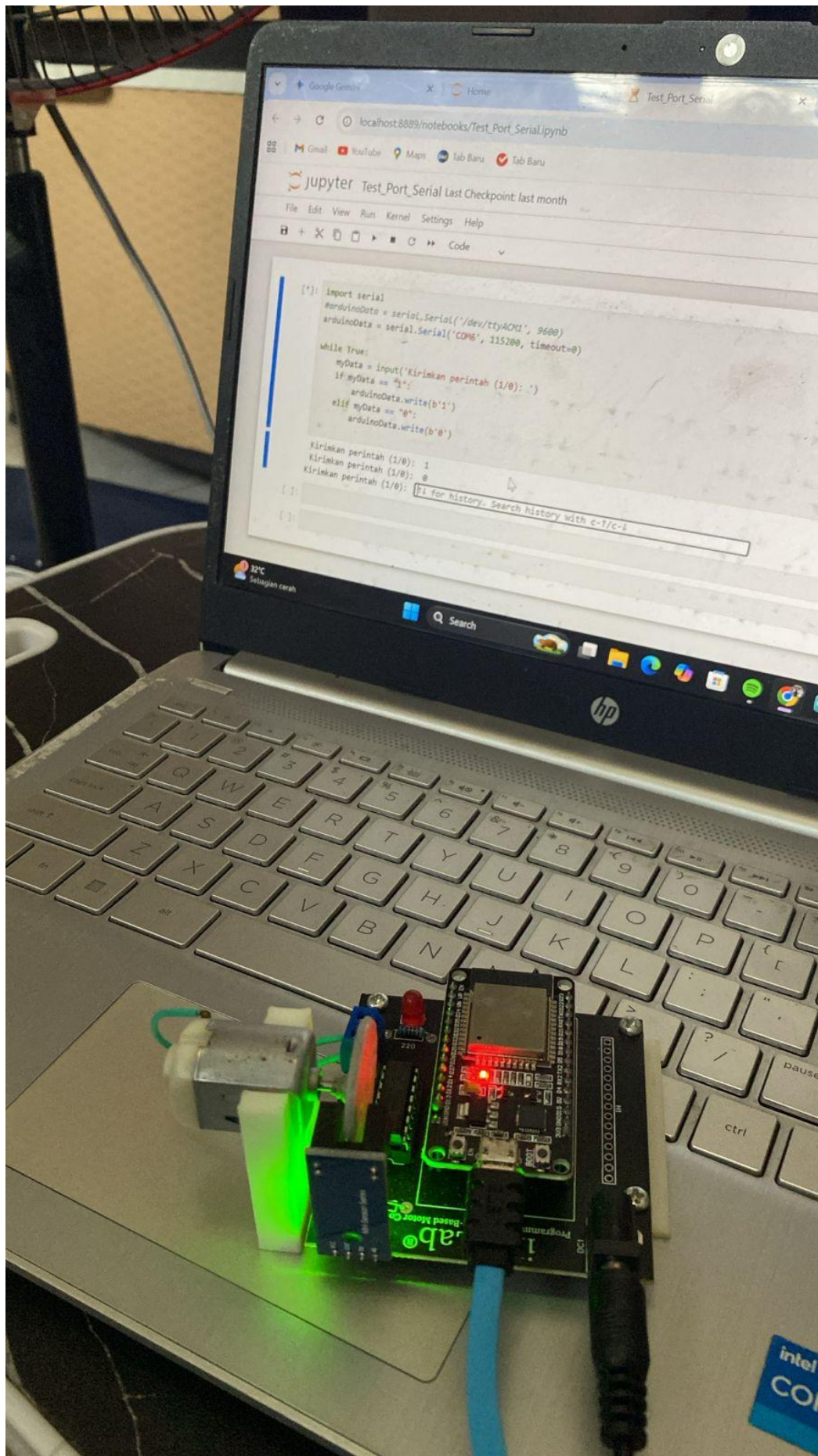
```
def version(self):  
    return "iMCLab Library v1.0 - UPN Veteran Jatim"  
  
def close(self):  
    if self.ser.is_open:  
        self.ser.close()  
        print("Arduino disconnected successfully")  
        return True  
    return False
```

Hasil Uji Lab Melalui Jupyter Pyhton

ON



OFF



Uji Kit IMCLab Melalui Hp Dengan IOT

Code IOT IMCLab

```
#include <WiFi.h>
#include <PubSubClient.h>
#include <Arduino.h>

int ledPin = 2;
char myData = 0;

int motor1Pin1 = 27;
int motor1Pin2 = 26;
int enable1Pin = 12;

// Setting PWM properties
const int freq = 30000;
const int pwmChannel = 0;
const int resolution = 8;
int dutyCycle = 200;

const byte pin_rpm = 13;
int volatile rev = 0;
//int rpm = 0;

const char* ssid = "PROBORINI"; // Enter your WiFi name
const char* password = "sudirjop"; // Enter WiFi password

#define mqttServer "broker.emqx.io"
#define mqttPort 1883

WiFiServer server(80);
WiFiClient espClient;
PubSubClient client(espClient);

String Topic;
String Payload;

// constants
const int baud = 115200;    // serial baud rate

void setup()
{
  pinMode(ledPin, OUTPUT);
  Serial.begin(115200);
  // sets the pins
  pinMode(motor1Pin1, OUTPUT);
  pinMode(motor1Pin2, OUTPUT);
  pinMode(enable1Pin, OUTPUT);
  pinMode(pin_rpm, INPUT_PULLUP);
  // pinMode(pin_rpm, INPUT);
}
```

```

//attachInterrupt(digitalPinToInterrupt(pin_rpm), isr, RISING);

// configure LED PWM functionalitites
ledcSetup(pwmChannel, freq, resolution);

// attach the channel to the GPIO to be controlled
ledcAttachPin(enable1Pin, pwmChannel);

// testing
Serial.print("Testing DC Motor...");

// Connect to WiFi network
Serial.println();
Serial.println();
Serial.print("Connecting to ");
Serial.println(ssid);
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected");

// Connect to Server IoT (CloudMQTT)
client.setServer(mqttServer, mqttPort);
client.setCallback(receivedCallback);

while (!client.connected()) {
    Serial.println("Connecting to CCloud IoT ...");

    // if (client.connect("ESP32Client", mqttUser, mqttPassword )) {
    if (client.connect("iMCLab On/Off")) {

        Serial.println("connected");
        Serial.print("Message received: ");

    } else {
        Serial.print("failed with state ");
        Serial.print(client.state());
        delay(2000);
    }
    client.subscribe("kontrolmyy");
}

}

void MotorOn()
{
    // Move DC motor forward with increasing speed

```

```

dutyCycle = 205;
digitalWrite(motor1Pin1, HIGH);
digitalWrite(motor1Pin2, LOW);
ledcWrite(pwmChannel, dutyCycle);
}

void MotorOff()
{
  dutyCycle = 0;
  digitalWrite(motor1Pin1, HIGH);
  digitalWrite(motor1Pin2, LOW);
  ledcWrite(pwmChannel, dutyCycle);
}

void receivedCallback(char* topic, byte* payload, unsigned int length) {

  /* we got '1' -> MotorOn() */
  if ((char)payload[0] == '1') {
    MotorOn();
    Serial.println("Motor On");
  }

  /* we got '0' -> Motoroff */
  if ((char)payload[0] == '0') {
    MotorOff();
    Serial.println("Motor Off");
  }
}

void loop()
{

  client.loop();

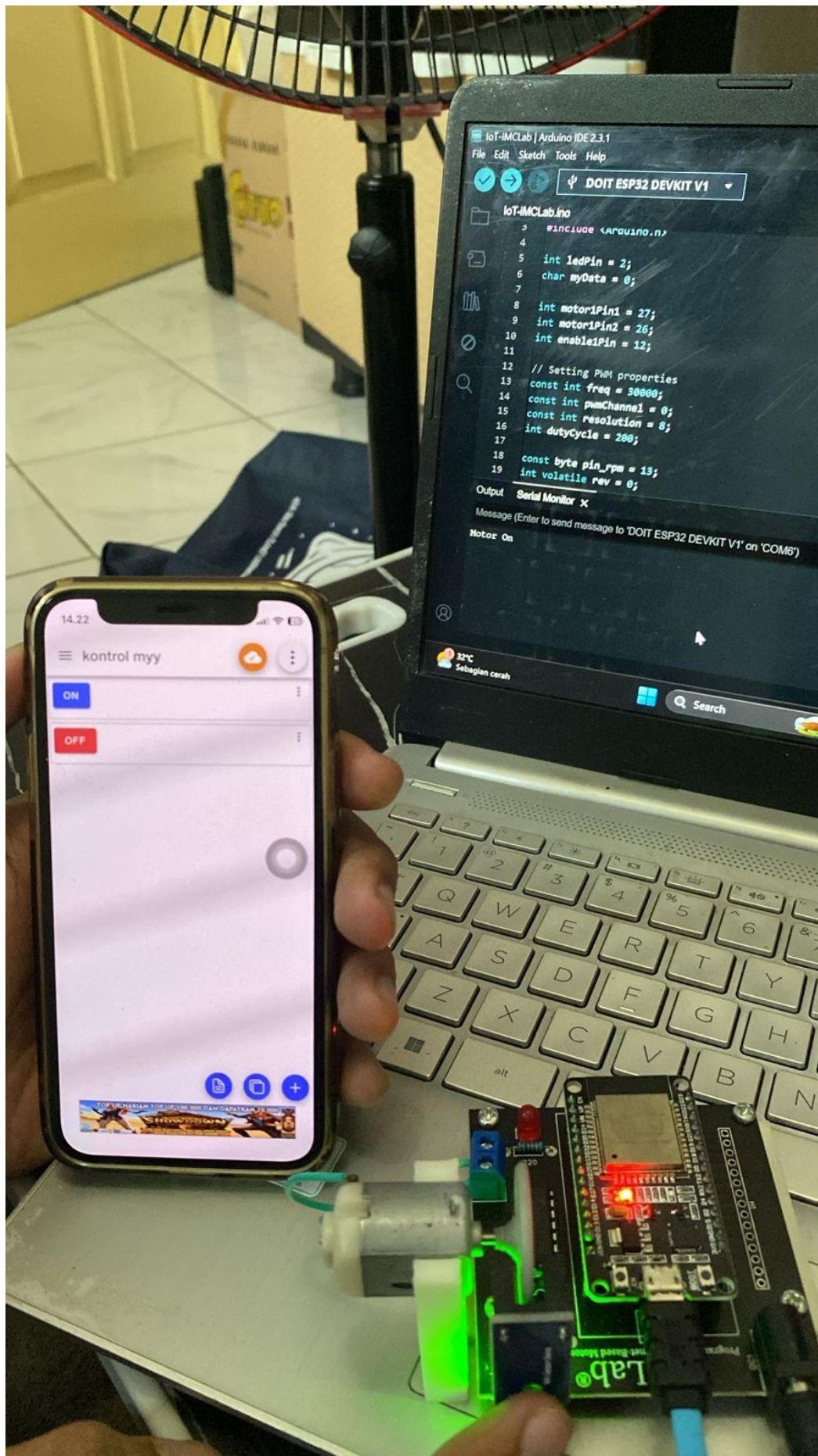
  myData = int(Serial.read());

  if (myData == '1'){
    Serial.println("LED is on !!!");
    digitalWrite(ledPin, HIGH);
    MotorOn();
  }
  else if (myData == '0'){
    Serial.println("LED is off !!!");
    digitalWrite(ledPin, LOW);
    MotorOff();
  }
}

```


Hasil Uji IMCLab Melalui HP Dengan IOT

ON



OFF



